Pre-Analysis Plan (PAP)

1- Empirical Strategy

To causally determine the effect of having daughters on beliefs, political attitudes, and household gender equality we adopt an instrumental variable (IV) design, using the gender of the first child as an instrument for having at least one daughter. This approach helps address the endogeneity of family composition and parental preferences.

Justification of the IV Approach

The primary empirical challenge is that the decision to have children (and possibly their number) may be correlated with unobserved parental attitudes toward gender equality and politics. Using the gender of the first child as an instrument is advantageous because:

- **Quasi-random variation:** Conditional on controls, the gender of the first child is essentially random.
- **Exogeneity:** It is plausibly unrelated to unobserved parental attitudes.

The IV strategy helps isolate exogenous variation in the treatment (having a daughter), providing a causal interpretation of the estimates. Preliminary analysis of a pilot sample (5%) shows that the first-stage F-statistic for the instrument exceeds 10, suggesting that the instrument is not weak. We will report and discuss the first-stage F-statistics for all final specifications.

To mitigate residual confounding, we include relevant covariates (age, education, number of childr etc.). Controlling for the total number of children prevents bias from fertility stopping rules.

Following standard practice in the literature, we compute robust standard errors clustered at the household level, accounting for both heteroskedasticity and intra-household correlation. This is done because of non-independence of observations within households. In the SOEP, there may be multiple respondents from the same household across different waves (e.g., both parents or multiple years of responses). Standard OLS/2SLS assumes independent errors, but household members' attitudes or political preferences are likely correlated due to shared environment, norms, or events. Ignoring this correlation could lead to underestimated standard errors and overstated significance levels.

2- Data and Sample

Data Source: SOEP CORE

Period: Our analysis covers data from 2000 to 2020, with the most recent available variables drawn from 2018 for H1 and H4, and from 2019 for H2 and H3. We decided not to use data from 2020 to avoid potential bias introduced by shifts in attitudes during the COVID-19 pandemic.

Sample: We restrict our sample to respondents with at least one child.

Overview of exclusions (detailed under hypotheses): We include only individuals with no missing values in the dependent variable, in any variable used to construct the dependent variable, and in the key independent variables. For H3, we additionally restrict the sample to individuals with partners recorded in SOEP, and ensure the partners also have no missing data in the variables listed above.

2. Hypotheses and Models

H1. Having daughters leads to stronger support for gender equality (as measured by attitudes)

Variables

agreement_sharework

Description	Response to the survey question: "Best if man and woman work the same amount so they can share the responsibility." We select this item because it is a single-question measure that serves as a strong proxy for support for gender equality and is available in a recent data collection wave.
Usage	Dependent variable
Values	[1] Completely Disagree to [7] Fully Agree
Generation	Taken from `pl/plh0308_v2`
Exclusion	This question does only exist in 2018 for the dataset we have. Thus we take 2018 values.

firstkid_female

Description	Binary variable indicating whether the first child is female.
Usage	Instrumental Variable
Categories	[1] Female, [0] Male
Generation	Generated dummy from `biobirth/kidsex01`.

has_daughters

Description	Binary variable indicating whether the respondent has at least one daughter.
Usage	Endogenous variable (instrumented by `firstkid_female`)

Categories	[1] Has at least one daughter, [0] Has no daughters
Generation	Generated from the SOEP `biobirth` module by checking the reported sex of all children (`kidsex01`to `kidsex19`).

num_biochildren

Description	Number of biological children reported by the respondent at the time of survey.
Usage	Control variable
Generation	Generated from SOEP `biobirth` by counting valid entries (`kidsex01`to `kidsex19`) for given person.
Exclusion	Limited to children born in 2018 or earlier to align with the timing of the dependent variable.

male

Description	Binary variable indicating whether the respondent is male.
Usage	Control variable
Categories	[1] Male, [0] Female
Generation	Generated from `ppathl/sex`
Exclusion	Restricted to survey year 2018 to align with the timing of the outcome variable

age

Description	Age of the respondent at the time of the survey.
Usage	Control variable
Generation	Calculated as survey year (e.g., 2018) minus respondent's year of birth (`ppathl/gebjahr`).
Exclusion	Restricted to survey year 2018 to align with the timing of the outcome variable

education_totalyears

Description	Total years of formal education completed by the respondent.
Usage	Control variable
Generation	Taken from `pgen/pgbilzeit`
Exclusion	Restricted to survey year 2018 to align with the timing of the dependent variable

logincome

Description	Log-transformed net monthly labor income of the respondent.
Usage	Control variable
Generation	Calculated as the natural logarithm of net labor income (`pgen/pglabnet`).
Exclusion	Restricted to survey year 2018 to align with the timing of the dependent variable

migback

Description	Migration background of respondent
Usage	Control variable
Categories	[0] No migration background, [1] Direct migration, [2] Indirect migration
Generation	Taken from `ppath/migback`

Exclusion	Restricted to survey year 2018 to align with the timing of the dependent variable

partner

Description	Categorical variable indicating the respondent's partnership status within the household.
Usage	Control variable
Categories	[0] No partner in household [1] Spouse or registered partner [2] Cohabiting (unmarried) partner
Generation	Taken from `ppath/partner`
Exclusion	Restricted to survey year 2018 to align with the timing of the dependent variable

years_since_firstkid

Description	Number of years since the respondent's first biological child was born.
Usage	Control variable
Generation	Calculated as surveey year (2018) minus the birth year of respondents first child. Taken from `biobirth/kidgeb01`

hid

Description	Unique household identifier
Usage	Clustering of standard errors at the household level.

Inclusion/Exclusion Criteria

In addition to variable-specific exclusions, we apply the following general exclusion criteria for the analysis:

- We include only respondents with at least one biological child born at least one year before the survey year (i.e., born in 2017 or earlier).
- We exclude individuals with missing values in the dependent variable, in any variable used to construct the dependent variable, or in key independent variables—namely sex, birth year, birth year of the first child, and number of children.

Analysis

We used an instrumental variables (IV) estimation strategy estimated through two-stage least squares (2SLS). Standard errors are be clustered at the household level to account for within-household correlation in the error terms.

The estimation proceeds as follows:

First stage:

$$HasDaughter_i = \alpha + \beta\, FirstChildFemale_i + \gamma' Controls_i + \varepsilon_i$$

Second stage:

$$agreement_sharework_i = \delta + \tau\, \widehat{HasDaugter}_i + \eta' Controls_i + u_i$$

The exact specification in Stata is as follows:

```
ivregress 2sls agreement_sharework (has_daughters = firstkid_female)
num_biochildren male age education_totalyears logincome i.migback
i.partner years_since_firstkid, cluster(hid)
```

Interpretation of Results

We choose the dependent variable agreement_sharework as a proxy for support for gender equality. We will interpret the coefficient τ on instrumented has_daugthers as the causal effect on having daughter on support for gender equality. H1 will be supported if $\tau > 0$ and $p < 0.05$.

H2. Having daughters leads to more left-leaning political preferences

Our analysis for H2 is similar to that of H1, except that the dependent variable is `political_attitude`, and control variables are constructed based on the 2019 survey year instead of 2018.

Variables

political_attitude

Description	Response to the survey question: "*In politics people often talk about "left" and "right" when it comes to characterize different political attitudes. If you think about your own political views: Where would you place yours?*" We select this item because it reflects the research question well.
Usage	Dependent variable
Values	[0] far left to [10] far right
Generation	Taken from `pl/plh0004`
Exclusion	This question does only exist a few times in our data and we take the most recent wave in 2019.

firstkid_female

Description	Binary variable indicating whether the first child is female.
Usage	Instrumental Variable
Categories	[1] Female, [0] Male
Generation	Generated dummy from `biobirth/kidsex01`.

has_daughters

Description	Binary variable indicating whether the respondent has at least one daughter.
Usage	Endogenous variable (instrumented by `firstkid female`)

Categories	[1] Has at least one daughter, [0] Has no daughters
Generation	Generated from the SOEP `biobirth` module by checking the reported sex of all children (`kidsex01`to `kidsex19`).

num_biochildren

Description	Number of biological children reported by the respondent at the time of survey.
Usage	Control variable
Generation	Generated from SOEP `biobirth` by counting valid entries (`kidsex01`to `kidsex19`) for given person.
Exclusion	Limited to children born in 2019 or earlier to align with the timing of the dependent variable.

male

Description	Binary variable indicating whether the respondent is male.
Usage	Control variable
Categories	[1] Male, [0] Female
Generation	Generated from `ppathl/sex`
Exclusion	Restricted to survey year 2019 to align with the timing of the outcome variable

age

Description	Age of the respondent at the time of the survey.
Usage	Control variable
Generation	Calculated as survey year (e.g., 2019) minus respondent's year of birth (`ppathl/gebjahr`).
Exclusion	Restricted to survey year 2019 to align with the timing of the outcome variable

education_totalyears

Description	Total years of formal education completed by the respondent.
Usage	Control variable
Generation	Taken from `pgen/pgbilzeit`
Exclusion	Restricted to survey year 2019 to align with the timing of the dependent variable

logincome

Description	Log-transformed net monthly labor income of the respondent.
Usage	Control variable
Generation	Calculated as the natural logarithm of net labor income (`pgen/pglabnet`).
Exclusion	Restricted to survey year 2019 to align with the timing of the dependent variable

migback

Description	Migration background of respondent
Usage	Control variable
Categories	[0] No migration background, [1] Direct migration, [2] Indirect migration
Generation	Taken from `ppath/migback`
Exclusion	Restricted to survey year 2019 to align with the timing of the dependent variable

partner

Description	Categorical variable indicating the respondent's partnership status within the household.
Usage	Control variable
Categories	[0] No partner in household [1] Spouse or registered partner [2] Cohabiting (unmarried) partner
Generation	Taken from `ppath/partner`
Exclusion	Restricted to survey year 2019 to align with the timing of the dependent variable

years_since_firstkid

Description	Number of years since the respondent's first biological child was born.
Usage	Control variable
Generation	Calculated as survey year (2019) minus the birth year of respondents first child. Taken from `biobirth/kidgeb01`

hid

Description	Unique household identifier
Usage	Clustering of standard errors at the household level.

Inclusion/Exclusion Criteria

In addition to variable-specific exclusions, we apply the following general exclusion criteria for the analysis:

- We include only respondents with at least one biological child born at least one year before the survey year (i.e., born in 2018 or earlier).
- We exclude individuals with missing values in the dependent variable, in any variable used to construct the dependent variable, or in key independent variables—namely sex, birth year, birth year of the first child, and number of children.

Analysis

We used an instrumental variables (IV) estimation strategy estimated through two-stage least squares (2SLS). Standard errors are be clustered at the household level to account for within-household correlation in the error terms.

The estimation proceeds as follows:

First stage:

$$HasDaughter_i = \alpha + \beta\, FirstChildFemale_i + \gamma' Controls_i + \varepsilon_i$$

Second stage:

$$political_attitude_i = \delta + \tau\, \widehat{HasDaugter}_i + \eta' Controls_i + u_i$$

The exact specification in Stata is as follows:

```
ivregress 2sls political_attitude (has_daughters = firstkid_female)
num_biochildren male age education_totalyears logincome i.migback
i.partner years_since_firstkid, cluster(hid)
```

Interpretation of Results

We choose the dependent variable political_attitude which indicates left leaning political preferences (Smaller values indicate more left-leaning political preferences). We will interpret the coefficient τ on instrumented has_daugthers as the causal effect on having daughter on left-leaning political prefences. H1 will be supported if $\tau < 0$ and $p < 0.05$.

H3. Having daughters leads to more gender equality in the household (as measured by behaviors).

This analysis is meaningful in the context of household work; thus we restricted our analysis to couples. We constructed the variable diff_kids_hours, defined as the absolute value of the difference between the respondent's and their partner's reported number of hours spent on childcare during a typical working day. We use this measure as a proxy for gender equality in household behavior, under the assumption that greater equality corresponds to smaller differences in childcare time between partners. Our analysis for H3 is similar to that of H1 for the method and the controls.

Variables

diff_kids_hours

Description	Absolute difference in the number of hours spent on childcare between the respondent and their partner on a typical working day.
Usage	Dependent variable
Generation	Constructed from `pl/pli0044_v3` by taking the absolute value of the difference between the respondent's and their partner's reported childcare hours. We matched partners of the respondents using `ppath/parid`
Exclusion	We take 2019 values of the survey answers being the latest value before COVID-19 pandemic (hence extensive home-stays) started.

firstkid_female

Description	Binary variable indicating whether the first child is female.
Usage	Instrumental Variable
Categories	[1] Female, [0] Male
Generation	Generated dummy from `biobirth/kidsex01`.
Exclusion	

has_daughters

Description	Binary variable indicating whether the respondent has at least one daughter.
Usage	Endogenous variable (instrumented by `firstkid_female`)
Categories	[1] Has at least one daughter, [0] Has no daughters
Generation	Generated from the SOEP `biobirth` module by checking the reported sex of all children (`kidsex01`to `kidsex19`).
Exclusion	

num_biochildren

Description	Number of biological children reported by the respondent at the time of survey.
Usage	Control variable
Generation	Generated from SOEP `biobirth` by counting valid entries (`kidsex01`to `kidsex19`) for given person.
Exclusion	Limited to children born in 2019 or earlier to align with the timing of the dependent variable.

male

Description	Binary variable indicating whether the respondent is male.
Usage	Control variable
Categories	[1] Male, [0] Female
Generation	Generated from `ppathl/sex`
Exclusion	Restricted to survey year 2019 to align with the timing of the outcome variable

age

Description	Age of the respondent at the time of the survey.
Usage	Control variable
Generation	Calculated as survey year (e.g., 2019) minus respondent's year of birth (`ppathl/gebjahr`).
Exclusion	Restricted to survey year 2010 to align with the timing of the outcome variable

education_totalyears

Description	Total years of formal education completed by the respondent.
Usage	Control variable
Generation	Taken from `pgen/pgbilzeit`
Exclusion	Restricted to survey year 2010 to align with the timing of the dependent variable

logincome

Description	Log-transformed net monthly labor income of the respondent.
Usage	Control variable
Generation	Calculated as the natural logarithm of net labor income (`pgen/pglabnet`).
Exclusion	Restricted to survey year 2010 to align with the timing of the dependent variable

migback

Description	Migration background of respondent

Usage	Control variable
Categories	[0] No migration background, [1] Direct migration, [2] Indirect migration
Generation	Taken from `ppath/migback`
Exclusion	Restricted to survey year 2019 to align with the timing of the dependent variable

partner

Description	Categorical variable indicating the respondent's partnership status within the household.
Usage	Control variable
Categories	[0] No partner in household [1] Spouse or registered partner [2] Cohabiting (unmarried) partner
Generation	Taken from `ppath/partner`
Exclusion	Restricted to survey year 2019 to align with the timing of the dependent variable

years_since_firstkid

Description	Number of years since the respondent's first biological child was born.
Usage	Control variable
Generation	Calculated as survey year (2019) minus the birth year of respondents first child. Taken from `biobirth/kidgeb01`

hid

Description	Unique household identifier
Usage	Clustering of standard errors at the household level.

parid

Description	Partner identifier
Usage	Used to match partners' number of hours spent in the childwork at a working day (`pl/pli0044_v3`)
Exclusion	Restricted to respondents who have a partner at the time of the survey [2019]

Inclusion/Exclusion Criteria

In addition to variable-specific exclusions, we apply the following general exclusion criteria for the analysis:

- We restrict the analysis to couples living in the same household at the time of the survey (2019). Specifically, we include only individuals with a valid partner identifier.
- We exclude individuals if either their own or their partner's reported hours spent on childcare during a typical working day are missing.

- We further restrict the sample to individuals whose youngest biological child was born at least two years prior to the survey year, to minimize potential bias from intensive maternal caregiving.
- We do not exclude individuals on extended parental leave, as such leave-taking reflects meaningful behavioral choices relevant to our analysis.
- We include only respondents with at least one biological child born at least one year before the survey year (i.e., born in 2018 or earlier).
- We exclude individuals with missing values in the dependent variable, in any variable used to construct the dependent variable, or in key independent variables—namely sex, birth year, birth year of the first child, and number of children.

Analysis

We used an instrumental variables (IV) estimation strategy estimated through two-stage least squares (2SLS). Standard errors are be clustered at the household level to account for within-household correlation in the error terms.

The estimation proceeds as follows:

First stage:

$$HasDaughter_i = \alpha + \beta\, FirstChildFemale_i + \gamma' Controls_i + \varepsilon_i$$

Second stage:

$$diff_kids_hours_i = \delta + \tau\, \widehat{HasDaugter}_i + \eta' Controls_i + u_i$$

The exact specification in Stata is as follows:

```
ivregress 2sls diff_kids_hours (has_daughters = firstkid_female)  male
age education_totalyears logincome i.migback i.partner
years_since_firstkid, cluster(hid)
```

Interpretation of Results

We choose the dependent variable diff_kids_hours as a (negative) proxy for gender equality in household. We will interpret the coefficient τ on instrumented has_daugthers as the causal effect on having daughter on support for gender equality. H3 will be supported if $\tau < 0$ and $p < 0.05$.

H4. Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes)

Our analysis for H4 is similar to that of H1, except that we differentiate the causal effect of having daughters by respondent gender. To do this, we modify the regression specification to include an interaction between the endogenous variable `has_daughters` and a binary indicator for male respondents (`male`).

Variables

agreement_sharework

Description	Response to the survey question: "Best if man and woman work the same amount so they can share the responsibility." We select this item because it is a single-question measure that serves as a strong proxy for support for gender equality and is available in a recent data collection wave.
Usage	Dependent variable
Values	[1] Completely Disagree to [7] Fully Agree
Generation	Taken from `pl/plh0308_v2`
Exclusion	This question does only exist in 2018 for the dataset we have. Thus we take 2018 values.

firstkid_female

Description	Binary variable indicating whether the first child is female.
Usage	Instrumental Variable
Categories	[1] Female, [0] Male
Generation	Generated dummy from `biobirth/kidsex01`.

has_daughters

Description	Binary variable indicating whether the respondent has at least one daughter.
Usage	Endogenous variable (instrumented by `firstkid_female`)
Categories	[1] Has at least one daughter, [0] Has no daughters
Generation	Generated from the SOEP `biobirth` module by checking the reported sex of all children (`kidsex01`to `kidsex19`).

has_daughters_male

Description	Interaction term between `has_daughters` and `male.`
Usage	Endogenous variable (instrumented by `firstkid_female_male`)
Generation	Generated as the product of `has_daughters` and `male.`

firstkid_female_male

Description	Interaction term between `firstkid_female` and `male.`

Usage	Instrument for the endogenous interaction term has_daughters_male
Generation	Generated as the product of firstkid_female and male.

num_biochildren

Description	Number of biological children reported by the respondent at the time of survey.
Usage	Control variable
Generation	Generated from SOEP `biobirth` by counting valid entries (`kidsex01`to `kidsex19`) for given person.
Exclusion	Limited to children born in 2018 or earlier to align with the timing of the dependent variable.

male

Description	Binary variable indicating whether the respondent is male.
Usage	Control variable
Categories	[1] Male, [0] Female
Generation	Generated from `ppathl/sex`
Exclusion	Restricted to survey year 2018 to align with the timing of the outcome variable

age

Description	Age of the respondent at the time of the survey.
Usage	Control variable
Generation	Calculated as survey year (e.g., 2018) minus respondent's year of birth (`ppathl/gebjahr`).
Exclusion	Restricted to survey year 2018 to align with the timing of the outcome variable

education_totalyears

Description	Total years of formal education completed by the respondent.
Usage	Control variable
Generation	Taken from `pgen/pgbilzeit`
Exclusion	Restricted to survey year 2018 to align with the timing of the dependent variable

logincome

Description	Log-transformed net monthly labor income of the respondent.
Usage	Control variable
Generation	Calculated as the natural logarithm of net labor income (`pgen/pglabnet`).
Exclusion	Restricted to survey year 2018 to align with the timing of the dependent variable

migback

Description	Migration background of respondent
Usage	Control variable
Categories	[0] No migration background, [1] Direct migration, [2] Indirect migration
Generation	Taken from `ppath/migback`
Exclusion	Restricted to survey year 2018 to align with the timing of the dependent variable

partner

Description	Categorical variable indicating the respondent's partnership status within the household.
Usage	Control variable
Categories	[0] No partner in household [1] Spouse or registered partner [2] Cohabiting (unmarried) partner
Generation	Taken from `ppath/partner`
Exclusion	Restricted to survey year 2018 to align with the timing of the dependent variable

years_since_firstkid

Description	Number of years since the respondent's first biological child was born.
Usage	Control variable
Generation	Calculated as surveey year (2018) minus the birth year of respondents first child. Taken from `biobirth/kidgeb01`

hid

Description	Unique household identifier
Usage	Clustering of standard errors at the household level.

Inclusion/Exclusion Criteria

In addition to variable-specific exclusions, we apply the following general exclusion criteria for the analysis:

- We include only respondents with at least one biological child born at least one year before the survey year (i.e., born in 2017 or earlier).
- We exclude individuals with missing values in the dependent variable, in any variable used to construct the dependent variable, or in key independent variables—namely sex, birth year, birth year of the first child, and number of children.

Analysis

We used an instrumental variables (IV) estimation strategy estimated through two-stage least squares (2SLS). Standard errors are be clustered at the household level to account for within-household correlation in the error terms.

Since `has_daughters` is instrumented by the gender of the first child (`firstkid_female`), the interaction term (`has_daughters` × `male`) is also endogenous and must be instrumented accordingly. We therefore include an additional instrument: the interaction between `firstkid_female` and `male`.

The estimation proceeds as follows:

First stage:

$$HasDaughter_i \bullet Male_i = \alpha + \beta\,(FirstChildFemale_i \bullet Male_i) + \gamma' Controls_i + \varepsilon_i$$

Second stage:

$$AgreementSharework_i = \delta + \tau_1 \hat{HasDaugter}_i + \tau_2(\hat{HasDaughter}_i \bullet Male) + \eta' Controls_i + u_i$$

The exact specification in Stata is as follows:

```
ivregress 2sls agreement_sharework (has_daughters has_daughters_male =
firstkid_female firstkid_female_male)num_biochildren male age
education_totalyears logincome i.migback i.partner years_since_firstkid,
cluster(hid)
```

Interpretation of Results

We choose the dependent variable agreement_sharework as a proxy for support for gender equality. We will interpret the coefficient τ_2 on the instrumented interaction term `has_daugthers_male` as the differential causal effect on having daughters for men compared to women on support for gender equality. H1 will be supported if $\tau_2 > 0$ and $p < 0.05$.